

The Logical Theory of Information

By

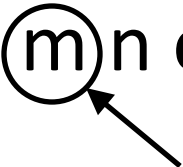
Robert Johannson

The Technical Definition of Information

The engineers at Bell Labs developed a simple definition of information:

Information is a message selected from a set of possible messages.

Possible messages:

a b c d e f g h i j k l  n o p q r s t u v w x y z

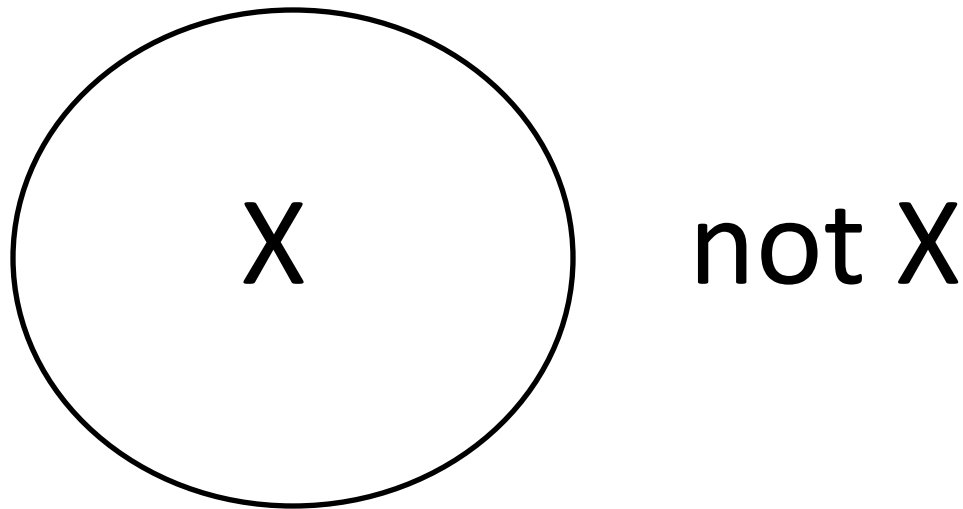
Information: the message chosen

The Basic Unit of Choice

The basic unit of choice is the binary choice: either/or.

This choice is the basis of Boolean logic.

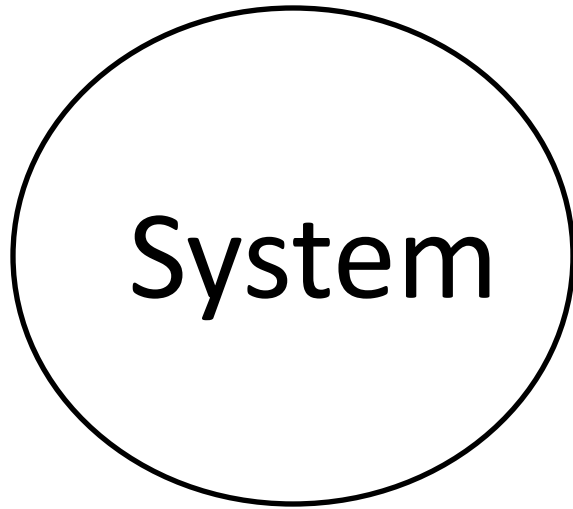
It is symbolized by the Venn diagram where the choice is represented by a line separating the x from the not x . It can also be used to diagram the syllogisms in traditional Aristotelian categorical logic.



Basic Binary Choice

Information and Systems

In Physics a system is defined by this basic binary choice. “A boundary is an imaginary line that separates the Universe into the system and its environment.” In other words systems are a function of information.



Environment

The Binary Code

Since the binary choice is the basic choice, any complex choice can be coded as a series of binary choices, usually symbolized by 0 and 1.

You can calculate the number of binary digits or “bits” needed to code a set of possibilities by taking the logarithm to the base 2 of the number of possibilities. Thus with 26 letters and a space we have $\text{Log}_2 27 = 5$ bits. Thus using the binary either/or as our basic unit of information we can count the amount of information by counting the number of bits it takes to code it.

Thus

A = 00000

B = 00001

C = 00010

D = 00011

E = 00100

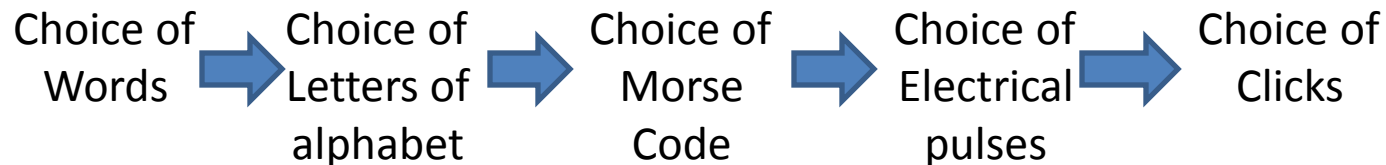
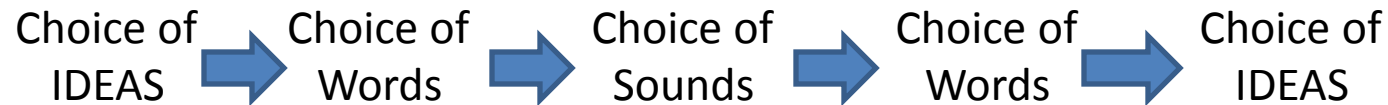
F = 00101

Etc.

This is the basis of modern computing where the computer is a chip with transistors that can be used as a gate either open or closed. Thus the binary code becomes a series of transistor positions. Computers are programmed with Boolean logic.

Communication as Translation

Since information is choice within a set, communication consists of translating choices in one set of possibilities into choices in another set, in other words coding.

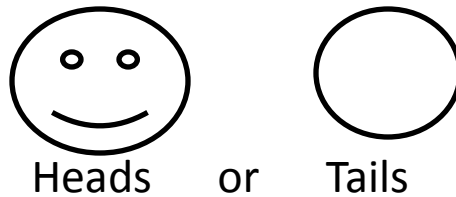


Although information consists of choices in a set and has no meaning as such. The meaning is in the reference. Thus the map is not the territory, but the territory is the meaning of the map. Thus the clicks mean electrical pulses which mean Morse code which mean letters which mean words which mean ideas.

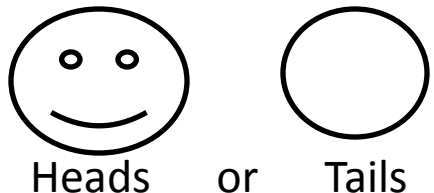
Possibilities and Probabilities

Since both information and statistical probabilities are calculated from a set of possibilities there is often confusion about the difference. Some people even think that information is a probability function. The difference is that information is measured in bits and probabilities are measured in fractions.

A coin has two possibilities.



Probabilities

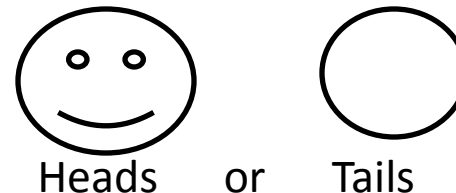


$$\frac{1}{2} = 0.5$$

$$\frac{1}{2} = 0.5$$

Probabilities are a fraction.

Information



1

Or

0

one bit

Information is measured in bits